databricks

Fewer empty shelves, before the doors open

Larkwell Foods · on-shelf availability



Databricks

Prepared for Larkwell Foods (fictitious) · Oct 2026

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More sales, store time back, and every gap explained

What Larkwell gets, with every number labelled by its source

Assumption

£23.7m

a year of sales recovered in the base case (range £15.8m to £39.5m)

Assumption

328,000

store hours a year given back: about £4.3m of capacity

Measured

98%

of gaps and lows get a confirmed cause automatically, every morning

Measured

Minutes

from the morning snapshot to action at the shelf, instead of the next day

About £40k to £90k a year to run · over 200× its cost in sales, and over 60× in gross profit — Estimate

Measured this build's data

Research published research

Assumption the pilot proves it

Estimate indicative list prices



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Larkwell Foods is fictitious · all data is synthetic

Every morning, £1.09m of fixable sales is at risk

Larkwell Foods: 900 stores, 6,000 lines and about £8.2bn of sales a year

Measured

205,834

shelves open empty or low, out of 5.4 million product-locations

Measured

£1.09m

of that day's sales at risk on the shelves Larkwell can fix

Measured + Research

£158m

a year of lost sales it could prevent: 40% of shoppers facing a gap buy elsewhere or not at all

The evidence sits in six systems

Stock	Deliveries	Orders
Stock counts	Waste	Allocations

Nobody sees them together, so the cause is found by hand, the next day, if at all.

Who feels it

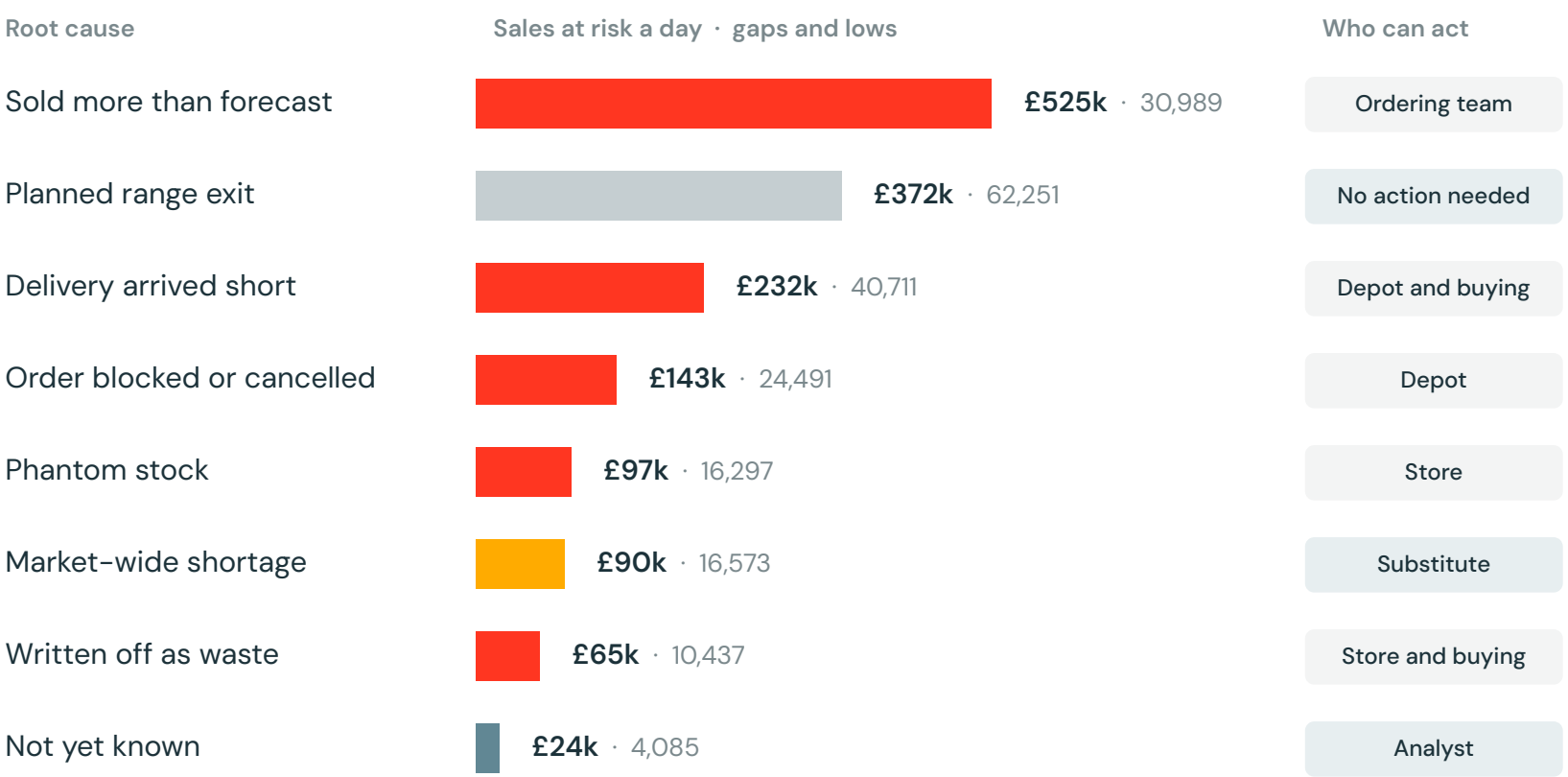
- **COO** lost sales, and store hours spent hunting for causes
- **Head of Availability** no evidence for supplier and depot conversations
- **Store colleague** an empty shelf, and no next step

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Most empty shelves are fixable; a third are deliberate

Gaps and lows by root cause: the sales at risk each day, and who can act



Measured

Business date 19 October 2026 · 205,834 gaps and lows

£525k a day

is the biggest fixable cause: lines selling faster than forecast. A forecasting problem, not a store one.

30%

of gaps and lows are planned range exits that teams still chase. They need no action at all.

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Four moments in Larkwell's day

What I'll show you in the app, and why each one matters

1 6:00 · head office

Head office sees what's wrong, and why

YOU'LL SEE

The dashboard and the drill-down, and the worst depot and supplier.

WHY IT MATTERS

98% of gaps explained before the first store opens.

2 In the aisle

A colleague fixes it at the shelf

YOU'LL SEE

Shelf Scan: the cause and the next action in milliseconds, and a one-tap action log.

WHY IT MATTERS

Action in seconds, at the shelf, not the next day.

3 A minute later

Anyone can ask, and the loop closes

YOU'LL SEE

Genie in plain English, and the action just logged, already governed.

WHY IT MATTERS

One number everyone trusts, and proof the work got done.

4 Before close

The store sees tomorrow's gaps today

YOU'LL SEE

The gap-risk model's watch list, with its reasons, and how yesterday's list did.

WHY IT MATTERS

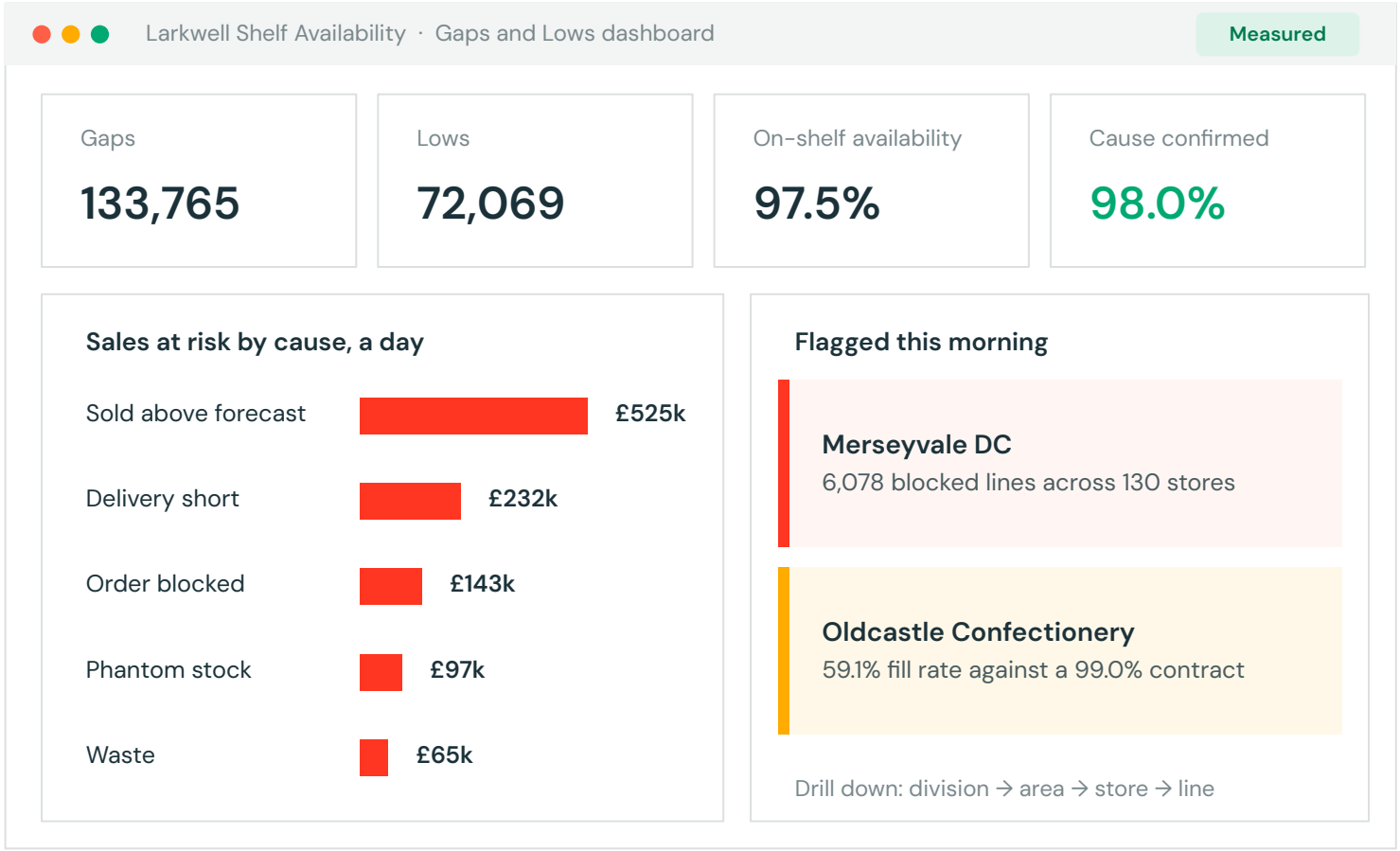
Act today on the shelves that will be empty tomorrow.

Then: how it's built, and what it's worth.

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6am: head office sees what's wrong, and why

Moment 1 · the Gaps and Lows dashboard, with this build's figures



YOU'LL SEE

- Every gap and low with its cause, by 6am
- A drill-down from division to the shelf
- The worst depot and supplier, with the evidence

WHY IT MATTERS

Teams stop arguing about what's wrong and start fixing why. The biggest fixable cause is a forecasting conversation, not a store one.

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At the shelf: the cause and the fix in milliseconds

Moment 2 · Shelf Scan at Covent Garden, with the app's real answer

1. **Scan**

   A colleague scans the shelf-edge label of an empty space.

2. **Look up** — 2.5 ms · Lakebase

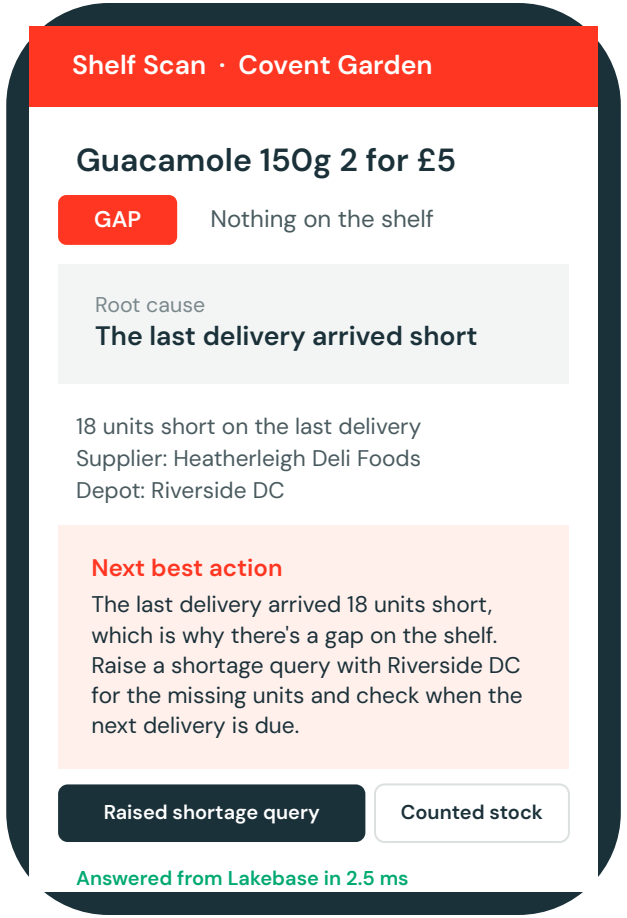
   One primary-key lookup in Lakebase, synced from the gold table.

3. **Advise** — async · LLM

   The playbook picks the action; the LLM words it for the colleague.

4. **Act** — 14 s · Unity Catalog

   One tap logs it, and Lakehouse Sync brings it back to Unity Catalog.



The playbook decides. The model words it.

```
# The rules pick the cause.
# The playbook picks the action:
ACTION_BY_REASON = {
  "SUPPLIER_SHORT":
    "Raise a shortage query...",
  "KNOWN_GAP":
    "No action needed...",
  # ...eight causes in all
}
# The LLM only words it.
```

- Every number must appear in the facts it's given
- It can't contradict the playbook: tested every day
- A fixed template answers if the model is down

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Anyone can ask, and the loop closes

Moment 3 · Genie on the metric view, and the action just logged

Ask the data · Genie

Which depot has the most order or delivery blocks?

Merseyvale DC has the most: 6,078 blocked lines across 130 stores.

Depot	Blocked lines	Stores
Merseyvale DC	6,078	130
Wealden DC	4,419	140
Riverside DC	3,848	175

Show SQL | Built on the metric view

What actions have stores logged today?

16 actions so far; 7 are shortage queries on lines that arrived short.

The loop closes, within seconds

A colleague taps “Raised shortage query”
Covent Garden · Guacamole 150g 2 for £5

▽

Written to Lakebase in milliseconds
Postgres: the app never writes to the lakehouse

▽

Lakehouse Sync streams the change
Every insert and update, with its history

▽

In Unity Catalog 14 seconds later
gold.colleague_actions: governed, with no personal data

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Before close: tomorrow's empty shelves, flagged today

Moment 4 · Covent Garden's briefing, from the gap-risk model

Larkwell Shelf Availability · Morning briefing · Covent Garden

Measured

Tomorrow morning

ML · ml.gap_risk v2

Predicted the day before, for 20 Oct. Tested on a week it never saw: 60% of each store's top 25 were empty the next morning, against 46% for “what's empty or low today”. Here, yesterday's list: 14 of 25 empty this morning.

Look fine now, likely to be empty tomorrow: worth a check before close

Product	Chance	Why
Orange Squash 1L	12%	Empty on 6 of the last 7 days
Organic Feta 200g	12%	Empty yesterday, refilled today
Coronation Chicken Sandwich 2pk Family Pack	11%	Empty on 2 of the last 7 days
Whole Milk 4 pints Family Pack	10%	Empty on 3 of the last 7 days
Organic Wholemeal Loaf 400g	9%	Shelf only 33% full; Empty yesterday, refilled today

Fix first: Easy Peeler Clementines 600g Multi-Buy (gap now, 77% likely still empty tomorrow) and 4 more of today's gaps and lows.

HOW IT WORKS

1. Lakeflow builds 34 features per line, every day
2. LightGBM trained in MLflow, tested on a later week
3. Champion in Unity Catalog only if it beats both rules
4. Scored every day for tomorrow; graded every morning

WHY IT MATTERS

Stores act today on the shelves that will be empty tomorrow. And the model is marked every day against what really happened.

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What changes on Monday morning

The same morning, for the three people who feel it

Store colleague

At the shelf

TODAY

An empty shelf, no reason, and no next step.

WITH THIS BUILD

Scans it: the cause and next action in seconds; before close, the shelves likely to be empty tomorrow.

KPI **Colleague minutes per check**

Head of Availability

Domain owner

TODAY

Causes found by hand, the next day. Supplier and depot calls without evidence.

WITH THIS BUILD

98% explained by 6am, with evidence for every supplier and depot, and every action visible.

KPI **Controllable gap rate · time to action**

COO

Executive sponsor

TODAY

£158m a year of preventable lost sales, and store hours spent hunting for causes.

WITH THIS BUILD

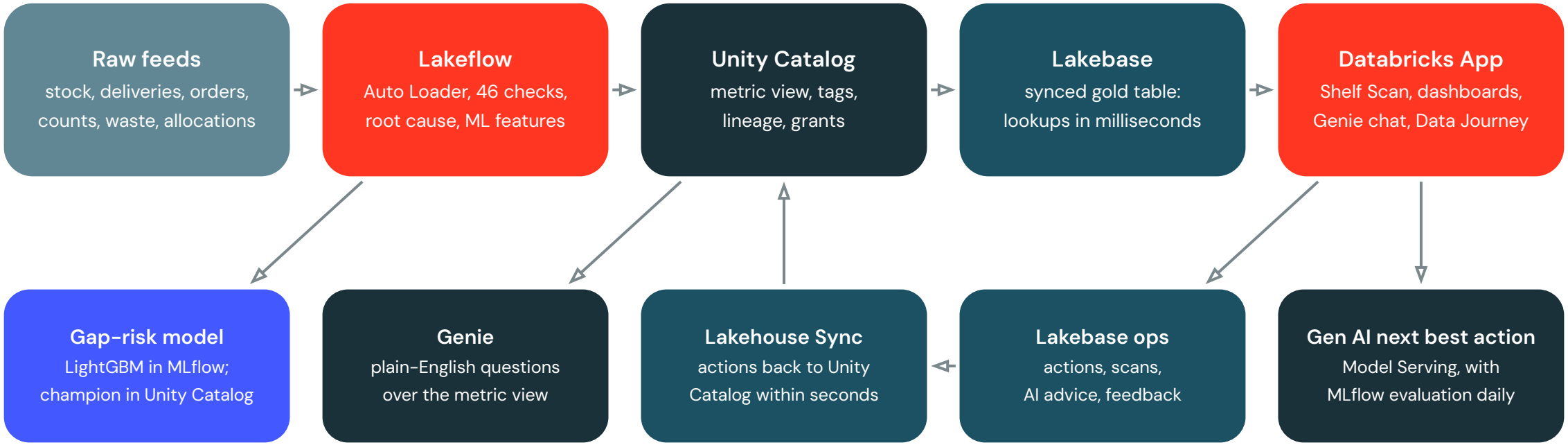
Fixable lost sales falling, store time back, and one number everyone trusts.

KPI **Sales recovered against control stores**

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One platform, end to end, checked every morning

Raw feeds to the shop floor and back, in one governed bundle



A Lakeflow Job runs the day: land → pipeline → Lakebase → 43 checks → predict gap risk, evaluate AI · one Declarative Automation Bundle deploys it all

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Every number is tested before anyone sees it

Execution evidence from the live workspace, kept as text in the repo

Measured

205,834

of 205,834 gaps and lows match an answer key the pipeline never reads, row by row

Measured

43 of 43

daily checks on every stage, from raw files to the model's inputs

Measured

72 of 72

end-to-end tests on the deployed app

Measured

2.5 vs 634 ms

a Shelf Scan lookup on Lakebase, against the SQL warehouse

Measured

12 of 12

Genie benchmark questions answered correctly

Measured

100%

of AI advice passes all eight checks; the judges agree with 19 hand-labelled answers

Measured

60% vs 46%

of each store's top-25 watch list empty next morning, against the today rule, on an unseen week

Measured

12.4% vs 2.2%

of lines that looked fine but were on the list were empty next morning, against lowest stock first

Every failed run is kept in the evidence, with what it caught. No threshold was lowered to get a pass.

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£23.7m a year back, for under £90k a year to run

For the COO: the base case, and what's measured or assumed

Measured

£1.09m

a day of controllable sales at risk

×

Research

40%

of at-risk demand is lost to the retailer

× 364 =

Measured + Research

£158m

a year of lost sales Larkwell could prevent

×

Assumption

15%

prevented by faster, targeted action (10% to 25%)

=

Assumption

£23.7m

a year recovered in the base case (£15.8m to £39.5m)

Assumption

£5.9m

a year of gross profit in the base case, at a 25% margin

Assumption

£4.3m

of store time a year: 328,000 hours, counted as capacity, not headcount

Estimate

Over 200×

the run cost in sales, and over 60× in gross profit; about £40k to £90k a year to run

Research: Gruen, Corsten and Bharadwaj, Retail Out-of-Stocks (GMA, FMI and CIES, 2002): 31% of shoppers buy elsewhere and 9% don't buy. The assumptions are what the pilot tests.

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Every gap explained, every supplier accountable

For the Head of Availability: the KPIs that change from day one

KPI	Today		With this build	
Root cause confirmed	By hand, the next day	→	**98%, automatically, every morning**	Measured
Time to action	The next day's report	→	**Minutes, at the shelf**	Measured
Supplier accountability	Anecdotal	→	**Oldcastle Confectionery: 59.1% fill against a 99.0% contract**	Measured
Depot accountability	Anecdotal	→	**Merseyvale DC: 6,078 blocked lines across 130 stores**	Measured
Planned exits chased	Often	→	**Never: 30% of gaps are marked “no action needed”**	Measured

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From diagnose, to predict, to act

Diagnose and a first predict are built, on the same data; act comes next

Horizon 1 — Built · this prototype

Diagnose

- A confirmed cause on every gap and low by 6am
- Scan-to-why at the shelf, in milliseconds
- Every action logged and governed

KPI: time to diagnose, next day → minutes

Horizon 2 — Built · first version

Predict

- Every line scored every day for the next morning
- MLflow, a Unity Catalog champion, graded every day
- Next: real history, intraday feeds, store task lists

KPI: share of gaps prevented

Horizon 3 — Then

Act

- Agents draft supplier claims and order uplifts
- A person approves anything with money attached
- Colleague feedback sharpens the rules

KPI: share of fixes automated

Rules explain; ML predicts

The LLM words, never decides

Every AI change is evaluated

One governed data product

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Proven in 12 weeks, against control stores

A pilot measured on these KPIs, with exit criteria agreed up front

WEEKS 0 TO 2

Connect

Connect real stock, delivery and order feeds with Lakeflow; check the rules against two weeks of history.

PROVES

The rules match reality

WEEKS 3 TO 8

Pilot

20 stores against 20 matched control stores, with the app in colleagues' hands.

PROVES

Gap rate, time to action, colleague minutes

WEEKS 9 TO 12

Scale

Roll out by division, with supplier and depot scorecards.

PROVES

Value at scale

What we need from you your availability lead · two data engineers · access to the stock, delivery and order extracts

The ask: approve the 6-week pilot in 20 stores, measured against 20 control stores.

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Three things to take away

Fewer empty shelves, before the doors open

1

Diagnosis, not detection, is the bottleneck

A confirmed cause on 98% of gaps by 6am, and the next action at the shelf in milliseconds.

2

Rules explain, ML predicts, AI words it

Causes from governed SQL; tomorrow's risk from a model graded every morning; the LLM only words the playbook.

3

One platform, end to end

Lakeflow, Unity Catalog, Lakebase, ML, Gen AI, Genie and an App: one bundle, checked every morning.

The ask: approve the 20-store pilot, measured against 20 control stores.

Thank you. I'd love your questions.

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Appendix

Answers to likely questions, for either of you

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Under the hood, for the architect

What each Databricks product does in this build, with the detail behind it

Lakeflow

- Auto Loader: 10 feeds; a full refresh rebuilt all 36 days at once
- 27 SQL files: bronze, silver, gold and the ML features
- 46 expectations; exactly the 120 malformed lines dropped

Unity Catalog

- One catalog: medallion schemas and governed volumes
- A metric view: one definition of every KPI, for the app and Genie
- Governed tags (pii on email), lineage, a read-only app identity

Lakebase

- Gold synced to Postgres: 5.4M rows, keyed on store and line
- 2.5 ms lookups, against 634 ms on the warehouse
- The app writes to Postgres; Lakehouse Sync streams changes back

Machine learning

- LightGBM on 34 features the pipeline builds
- MLflow run; Unity Catalog model with a gated champion alias
- Scored every day, graded every morning

Gen AI

- Claude Sonnet 4.5 on Model Serving words the playbook action
- Daily MLflow evaluation: 48 cases, six code checks, two judges
- Judges on GPT-5.5, calibrated on 19 hand-labelled answers

Genie

- Over the metric view, gold, supplier performance and the actions views
- Text instructions and 9 example queries
- Benchmark: 11, 11, then 12 of 12

App and operations

- FastAPI and a single-page app on Databricks Apps
- One bundle: pipeline, three jobs and the app
- Daily job: land, pipeline, Lakebase, 43 checks, ML, AI evaluation

Testing

- An answer key matched row by row, every morning
- 43 daily checks; 72 of 72 end-to-end tests on the app
- Every failed run kept in the evidence

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Decisions and trade-offs

Each choice, why it was made, and what it costs

Decision	Why	Trade-off
Rules for today's cause; ML for tomorrow's risk	Rules: explainable, tested row by row. ML: ranks risk better than any rule	The model needs real history: the pilot retrains it
Features in the pipeline; labels joined at training	One definition for training and scoring; nothing from the future leaks in	Six weeks recomputed daily, for now
Lakebase for Shelf Scan, the warehouse for analytics	Point lookups for thousands of colleagues, over 100× faster	A second, governed, read-only copy
App writes go to Lakebase, not Delta	Transactional speed; the app can't write to the lakehouse	Lakehouse Sync is in Beta; the fallback is a federated read
The LLM words the advice; it never decides it	No invented numbers or causes	Less 'creative' advice, by design
Daily snapshot sync, not continuous	Stock arrives once a day, so continuous sync would cost more for nothing	Revisit when stock feeds go intraday
Synthetic data with an answer key	Proves the pipeline on every run	Real-data noise is validated in the pilot

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Risks, and how each is handled

Business risks first, then technical

Risk	How it's handled
Adoption in stores	It fits the handheld routine; one tap to log; the pilot measures use as a KPI
The value doesn't materialise	Measured against 20 control stores, with exit criteria agreed before week 3
Cost	Serverless and autoscaling, about £40k to £90k a year; a smaller model or caching cuts the LLM cost
Feed data quality	46 expectations and 43 daily checks fail loudly before anyone sees a number
Wrong AI advice	Playbook-driven; a daily evaluation with thresholds that fail the job; a template fallback
Security and privacy	Least-privilege grants; pii tagged; no personal data in Genie's views; a read-only app identity
Beta feature (Lakehouse Sync)	A federated read over the Lakebase catalog, behind the same views
Model learned on simulated history	It proves the method; the pilot retrains it on real history before anyone relies on it
A retrain makes it worse	Gates: it must beat both rules and the champion; it's graded every morning

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How the AI advice is tested, every day

The app's exact prompt, tested on 48 real cases; a miss fails the job

Check	Type	What it checks	Bar	Final run
Numbers grounded	Code	Every number appears in the facts given	98%	100%
Follows the playbook	Code	The advice contains the playbook action	95%	100%
No contradiction	Code	No forbidden advice, such as chasing a supplier for a planned exit	100%	100%
No internal codes	Code	No codes such as SUPPLIER_SHORT	100%	100%
Units only	Code	No invented units such as cases	100%	100%
Concise	Code	At most three sentences and 75 words	90%	100%
Plain English	LLM judge	Plain, friendly British English for a colleague	90%	100%
Consistent with the playbook	LLM judge	Doesn't contradict the recommended action	95%	100%
Judge calibration	Both judges	Agree with 19 hand-labelled good and bad answers	100%	100%

The judges run on GPT-5.5, a different model family from the advice model (Claude Sonnet 4.5). No threshold was lowered to get a pass.

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The gap-risk model, tested like everything else

Trained on three weeks, tested on the next, graded every morning

Test week, all 900 stores	Model	Today rule	Lowest cover
Ranks tomorrow's empty shelves first (AUC)	0.93	0.83	0.83
Top 25 per store: empty next morning	60%	46%	46%
Top 25: share of tomorrow's empties caught	16%	12%	12%
Looks fine today: AUC	0.86	0.66	0.66
Looks fine, top 25: empty next morning	12.4%	2.2%	2.2%

Measured

Test week 12 Oct to 18 Oct, never seen in training. Both rules use only what a store sees in its morning report.

Predicts
an unplanned empty shelf tomorrow morning, for every ranged line, the day before

Learns from
34 pipeline features; most weight on today's shelf, the last 7 days and how full it is

Trained
21 Sep to 11 Oct; LightGBM, 169 trees; logged to MLflow

Gated
beats both rules and the champion, is calibrated, and passes the data checks

Graded
every morning: top-25 lists 58% to 61% empty over the last 7 mornings

Learned on simulated history: this proves the method. The pilot retrains it on real history and measures the gaps it prevents.

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Security and governance

Least privilege by default; no personal data where Genie can see it

Read-only app identity

The app's service principal reads gold and ops only. It can't write to the lakehouse.

Lakebase roles

It reads the serving table, and reads, inserts and updates its own tables. No deletes.

Personal data

Emails sit in one restricted table, tagged pii. Every view built for Genie leaves them out.

Tags and lineage

Governed data_classification and domain tags; lineage captured automatically.

In production

On-behalf-of-user auth, so Unity Catalog row filters apply per colleague.

This build

A fictitious retailer and synthetic data; a scan blocks publishing on names, hosts, emails or tokens.

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Value model: inputs and sources

Every input labelled: measured, published research, or an assumption

Input	Value	Source
Forecast demand	£22.6m a day (about £8.2bn a year)	Measured
Controllable sales at risk	£1.09m a day, across 127,010 gaps and lows	Measured
Share of at-risk demand lost to the retailer	40% (31% buy elsewhere, 9% don't buy)	Research: Gruen, Corsten and Bharadwaj (2002)
Prevented by faster, targeted action	10%, 15% or 25%	Assumption: the pilot measures it
Gross margin	25%	Assumption
Manual checks per store per day	20, saving 3 minutes each, at £13 an hour	Assumption

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Run cost (indicative)

List prices; to confirm with the account team

Component	Assumption	Estimate a year
Lakeflow pipeline, daily job and the gap-risk model	Serverless, about 20 minutes a day, and a weekly retrain	Low thousands of pounds
Lakebase	Autoscaling 1 to 8 CU, about 2 CU on average	Under £10k
Databricks App	One medium app, always on	Under £10k
SQL warehouse	Shared serverless warehouse, business hours	£10k to £20k
LLM next best action	About 36,000 calls a day; a smaller model or caching cuts this	£10k to £40k
Total		**About £40k to £90k**

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Root-cause rules, in priority order

The first rule with evidence wins; no evidence goes to an analyst

#	Root cause	Evidence	Who acts
1	Planned range exit	The line is exiting the store's range	No action needed
2	Market-wide shortage	The allocation was cut for a market shortage	Substitute
3	Delivery arrived short	Delivered less than ordered	Depot and buying
4	Order blocked or cancelled	The latest order status is blocked or cancelled	Depot
5	Phantom stock	A stock count removed stock that wasn't there	Store
6	Waste	Units written off	Store and buying
7	Sold more than forecast	Sales above forecast	Ordering team
8	Not yet known	No evidence	Analyst

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